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(54) **METHOD FOR MANUFACTURING
ACOUSTICAL ELEMENTS**

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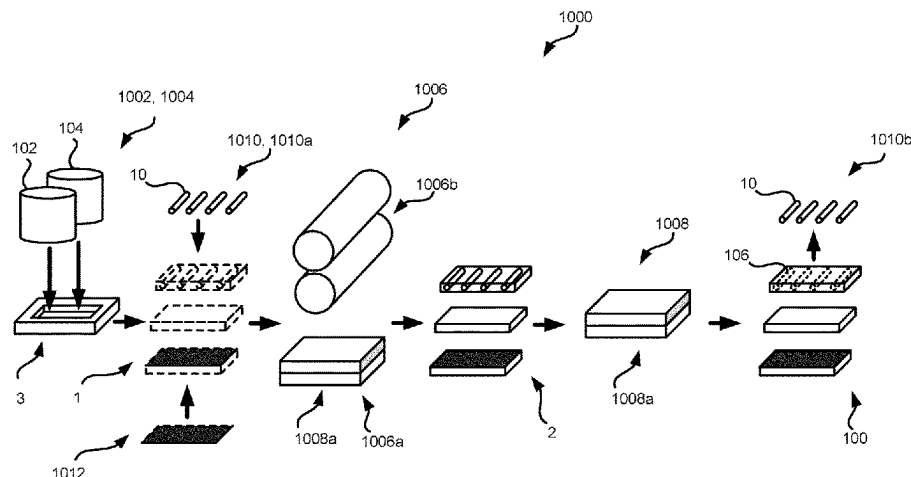
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ABSTRACT

A method for manufacturing acoustical elements, includes providing a first fibre component in a form of mineral wool and a second fibre component in a form of bicomponent fibres having a core with a thermoplastic outer layer, mixing the first fibre component and the second fibre component, for provision of a mixture, shaping the mixture into single layered tile shaped elements whereby the mixture is compressed with a compression ratio to a compressed state, and fixating the single layered tile shaped elements in the compressed state for obtaining the acoustical elements.

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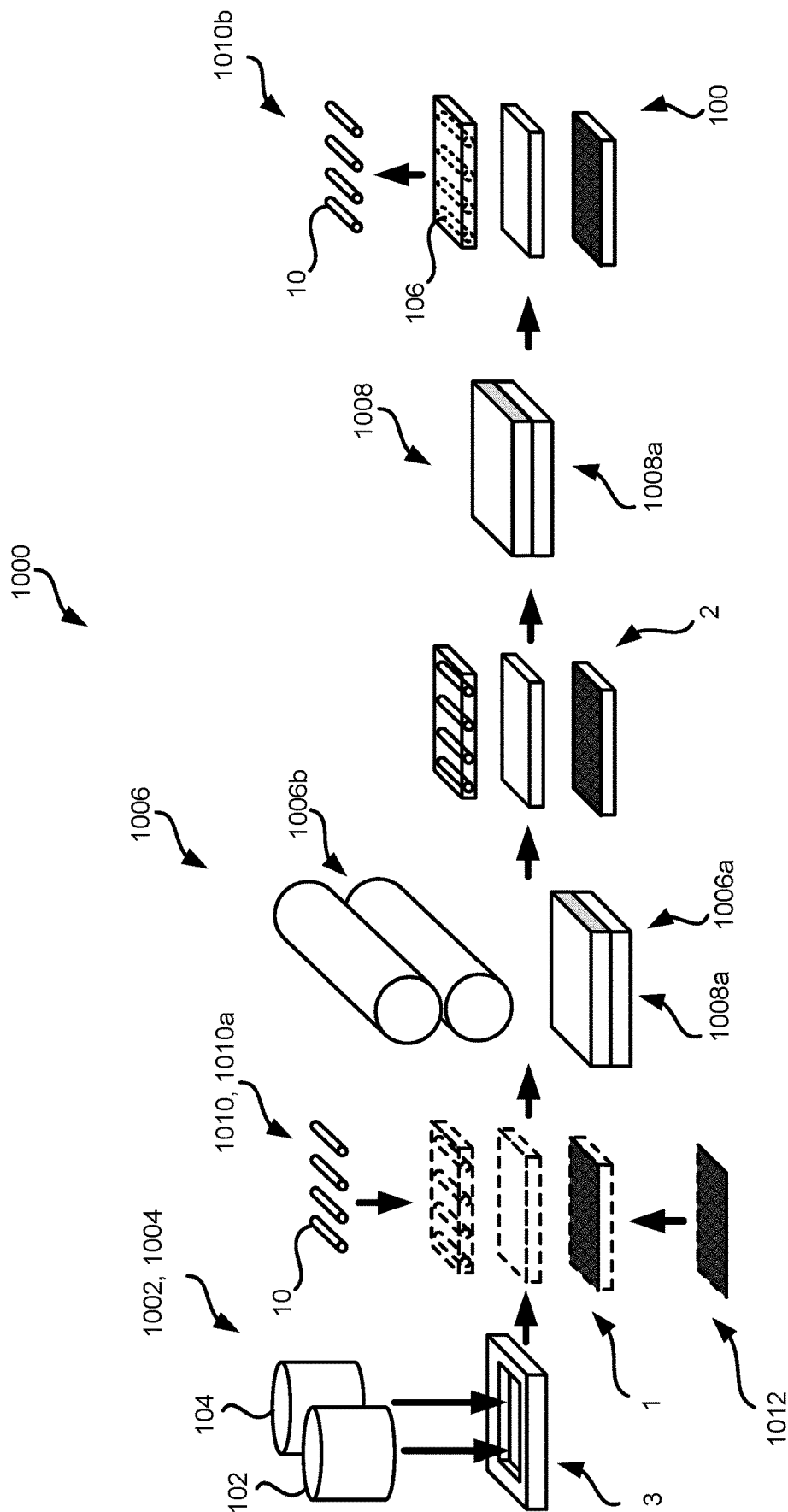
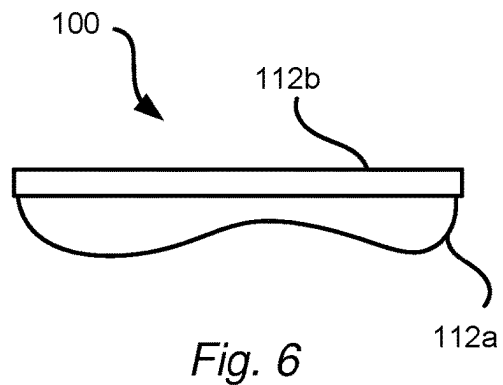
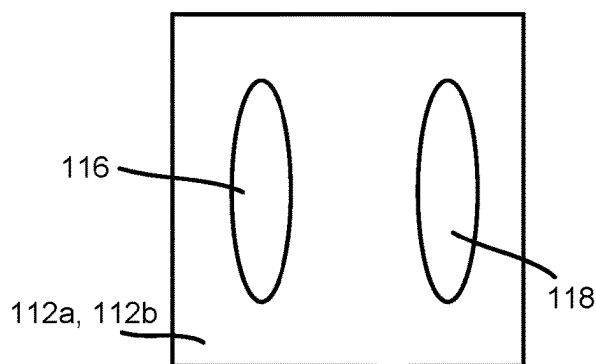
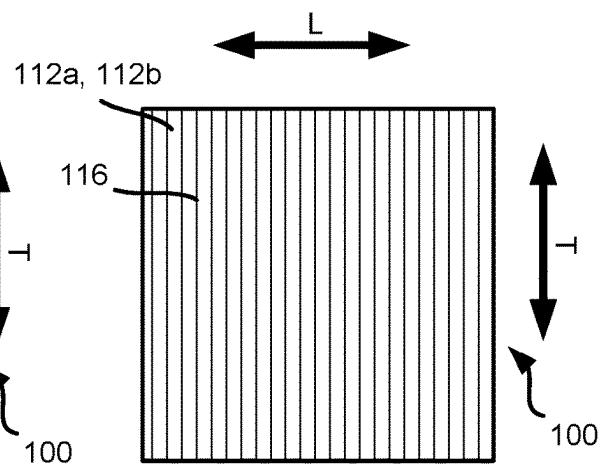
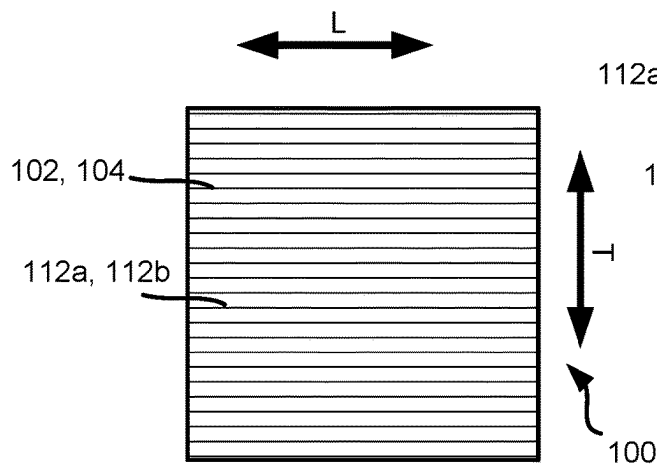
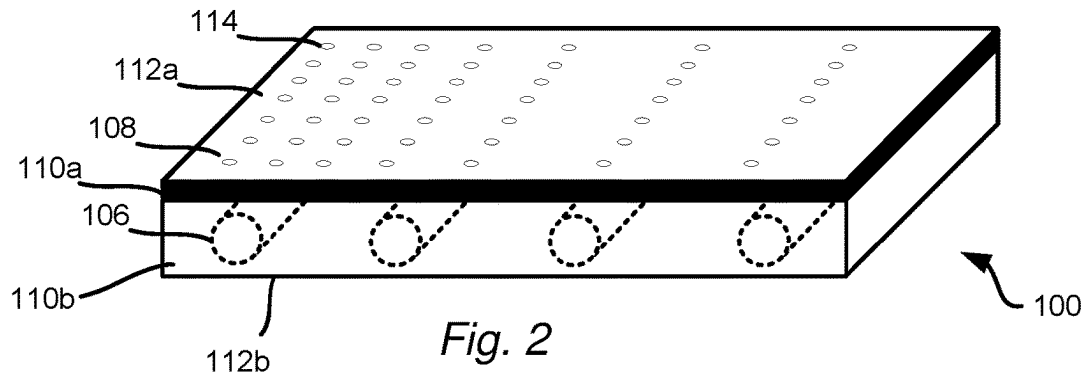
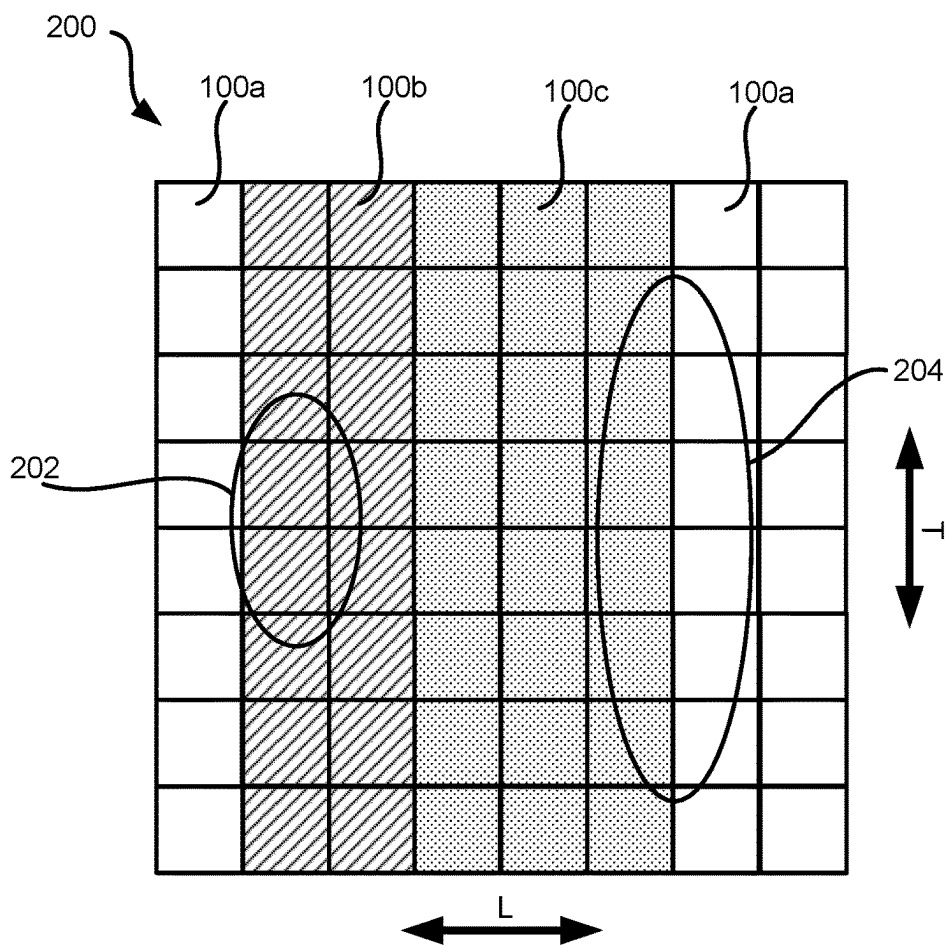
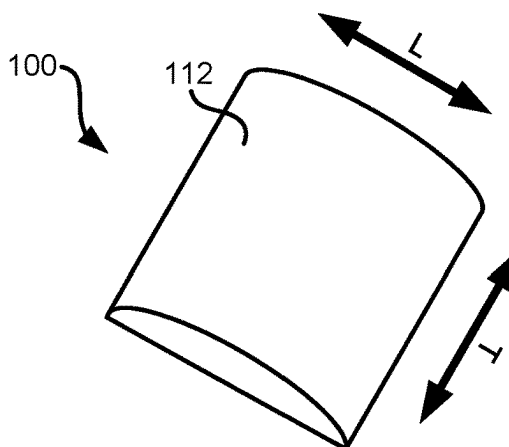
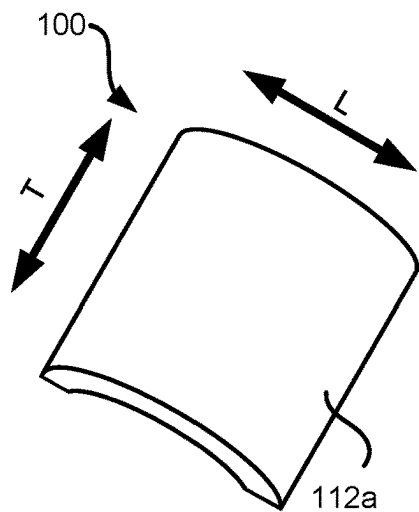


Fig. 1





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METHOD FOR MANUFACTURING ACOUSTICAL ELEMENTS

FIELD OF THE INVENTION

The present invention relates to acoustical elements, typically in the shape of ceiling tiles or wall panels. More specifically, the present invention relates to a method for manufacturing acoustical elements and to a suspended ceiling system comprising ceiling tiles in the form of acoustical elements manufactured according to the method.

BACKGROUND ART

Ceiling tiles in suspended ceiling systems serve several purposes. They should provide an aesthetically pleasing interior ceiling that allows covering of building equipment such as ventilation, ducts, and wiring. Some ceiling tiles form functional parts of the ventilation, for instance by being provided with nozzles for direct ventilation or by allowing air to be ventilated through or around the ceiling tiles to form diffuse ventilation.

In many cases, ceiling tiles are used to provide improved acoustics to the room in which they are mounted and may thus form acoustical elements. Such acoustical elements may also be mounted to the walls of a room. Many variations of the properties of acoustical elements are available, for instance the shape, density and air opacity may all be altered to achieve different acoustic properties. Different acoustical elements are thus available for achieving an acoustic environment in the room which suits the intended use. However, even if many different types of acoustical elements are available, it is generally difficult and time consuming to achieve combinations of ceiling tiles that provides the desired acoustics for each room.

Moreover, manufacturers of acoustical elements such as ceiling tiles generally strive towards reducing the environmental impact of their products. It is thus desired to find a way to provide acoustical elements that requires less resources than prior art acoustical elements.

SUMMARY OF THE INVENTION

In view of that stated above, the object of the teachings herein is to provide a method for manufacturing acoustical elements that alleviates some of the problems with prior art. It is also an object to provide a suspended ceiling system comprising acoustical elements manufactured according to the method.

To achieve at least one of the above objects and also other objects that will be evident from the following description, there is provided in a first aspect a method for manufacturing acoustical elements. The method comprises:

- providing a first fibre component in the form of mineral wool and a second fibre component in the form of bicomponent fibres having a core with a thermoplastic outer layer,
- mixing the first fibre component and the second fibre component, for provision of a mixture,
- shaping the mixture into tile single layered shaped elements whereby the mixture is compressed with a compression ratio to a compressed state, and
- fixating the single layered tile shaped elements in the compressed state for obtaining the acoustical elements, wherein fixating the single layered tile shaped elements comprises heating the single layered tile shaped elements in the compressed state for melting the thermo-

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plastic outer layer of the bicomponent fibres, whereby the second fibre component is activated for fixating the single layered tile shaped elements in the compressed state. Mixing the first fibre component and the second fibre component and shaping the mixture comprises controlling a mixing ratio of the first fibre component in relation to the second fibre component and/or the compression ratio of the mixture such that a first group of the acoustical elements having a first set of acoustic properties, such as sound absorbing properties, and a second group of the acoustical elements having a second set of acoustic properties being different from the first set of acoustical properties. The first fibre component preferably comprising a recycled fibre material in the form of grinded recycled ceiling tiles. The method allows adaptation and manufacturing of different groups of acoustical elements in one process, which facilitates provision of a suspended ceiling system that is adapted to the intended use whilst reducing environmental impact by using recycled materials. The manufacturing method of the acoustical elements can thus be adapted such that each acoustical element can be provided with acoustic properties that suits its intended use and position in e.g. a suspended ceiling system.

The shaping of the mixture may comprise forming the single layered tile shaped elements in a form pressing operation, such as vacuum forming, pressure forming and/or drape forming. Form pressing permits large variations of the shape and the compression ratio in different regions of each acoustical element.

Moreover, shaping the mixture may comprise a calendaring operation, providing additional variations to the surface topography and/or compression ratio of the acoustical element.

The mixing ratio may further be controlled such that the mixing ratio for a single acoustical element is non-uniform. Consequently, each acoustical element may comprise regions with varying proportions of the first and second fibre component. Each such region will exhibit different acoustical properties, allowing each acoustical element to be adapted for its intended use.

The compression ratio may further be controlled such that the compression ratio for a single acoustical element is non-uniform. The compression ratio could be varied such that a varying density of the acoustical element between 30 and 150 kg/m³ can be achieved.

Shaping the mixture may comprise forming single layered tile shaped elements with non-uniform thickness. Alternatively, the single layered tile shaped elements may be formed with a uniform thickness

The method may further comprise arranging core elements in the mixture before the step of shaping of the mixture and removing the core elements from the single layered tile shaped elements after the step of fixating of the single layered tile shaped elements for formation of cavities in the acoustical elements. The acoustical properties of the acoustical element can be further adapted by the provision of the cavities, each of which may alter the air flow resistance through the acoustical element and/or resonance properties throughout the fibre structure.

The core elements may be arranged for formation of a plurality of evenly distributed cavities in each acoustical element.

The shaping of the mixture into tile shaped elements may be performed such that the single layered tile shaped elements are provided with a single curved or double curved extension.

Shaping of the mixture into single layered tile shaped elements may be performed such that the single layered tile shaped elements obtain a front surface and/or rear surface provided with a topography, i.e. the front and/or rear surface being non-planar.

The method may further comprise providing each of the acoustical elements with a surface layer forming a front and/or back layer thereof, the surface layer provides additional possible variations to acoustical properties of each acoustical elements.

The surface layer may have a non-uniform air flow resistance.

In a second aspect is a suspended ceiling system provided comprising ceiling tiles in the form of acoustical elements manufactured by the method of the first aspect. The ceiling tiles are obtained from the first and second group of acoustical elements. The suspended ceiling system may thus be adapted to suit the intended use by provision of acoustical elements according to the manufacturing method disclosed herein in one manufacturing process.

Generally, all terms used in the claims are to be interpreted according to their ordinary meaning in the technical field, unless explicitly defined otherwise herein. All references to "a/an/the [element, device, component, means, step, etc]" are to be interpreted openly as referring to at least one instance of said element, device, component, means, step, etc., unless explicitly stated otherwise. The steps of any method disclosed herein do not have to be performed in the exact order disclosed, unless explicitly stated.

BRIEF DESCRIPTION OF THE DRAWINGS

The above, as well as additional objects, features and advantages of the present invention, will be better understood through the following illustrative and non-limiting detailed description of preferred embodiments of the present invention, with reference to the appended drawings, where the same reference numerals will be used for similar elements, wherein:

FIG. 1 discloses a schematic outline of a method for manufacturing acoustical elements.

FIG. 2 discloses a perspective view of an acoustical element manufactured according to the method disclosed herein.

FIG. 3 discloses an acoustical element manufactured according to the method disclosed herein.

FIG. 4 discloses an acoustical element manufactured according to the method disclosed herein.

FIG. 5 discloses an acoustical element manufactured according to the method disclosed herein.

FIG. 6 discloses a side view of an acoustical element manufactured according to the method disclosed herein.

FIG. 7 discloses a perspective view of an acoustical element manufactured according to the method disclosed herein.

FIG. 8 discloses a perspective view of an acoustical element manufactured according to the method disclosed herein.

FIG. 9 discloses a suspended ceiling system comprising ceiling tiles in the form of acoustical elements obtained from

a first group of acoustical elements and a second group of acoustical elements having different acoustic properties.

DESCRIPTION OF EMBODIMENTS

The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which currently preferred embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided for thoroughness and completeness, and fully convey the scope of the invention to the skilled person.

FIG. 1 shows a schematic outline of a method 1000 for manufacturing acoustical elements 100. The acoustical elements 100 are preferably tile shaped and configured to be arranged in a suspended ceiling system 200 (shown in FIG. 9). The acoustical elements 100 are intended to provide an aesthetically pleasing interior ceiling in a room in which the suspended ceiling 200 is mounted. Further, the acoustical elements 100 are intended to provide acoustic properties to the suspended ceiling system 200 that can be tailored for the desired use of the room in which the suspended ceiling system 200 is mounted. While the acoustical elements 100 in the present disclosure are described as being mounted to a suspended ceiling system, it is to be realized that the acoustical elements 100 could just as well be mounted to a wall in a room as wall panels.

Moreover, the method 1000 disclosed herein provides a way of utilizing recycled fibre material from older ceiling tiles.

The method 1000 comprises providing 1002 a first fibre component 102 in the form of mineral wool and a second fibre component 104 in the form of bicomponent fibres having a core with a thermoplastic outer layer. The first fibre component 102 may e.g. be grinded or shredded glasswool, preferably from recycled ceiling tiles that are grinded or shredded to form the first fibre component 102. The method 1000 may thus comprise grinding or shredding recycled ceiling tiles for forming the fibre component 102.

The second fibre component 104 is as mentioned bicomponent fibres. The bicomponent fibres have a core provided with a thermoplastic outer layer. The thermoplastic outer layer may be polyethylene, however other materials are also considered. The thermoplastic outer layer of the bicomponent fibres may thus form a binder that facilitates subsequent fixation of the two fibre components 1002, 104. Furthermore, a separate binder may be provided for fixation of the first and second fibre components 102, 104.

The first and second fibre components 102, 104 may further be provided in different proportions, i.e. different mixing ratios, which provides different acoustic properties to the acoustical elements 100.

The method 1000 further comprises mixing 1004 the first fibre component and the second fibre component 102, 104, for provision of a mixture 1. The mixing 1004 may be performed by forming a homogenous mixture 1 in which the first fibre component 102 and the second fibre component 104 are evenly distributed in the mixture 1. The mixing 1004 may be controlled such that the mixing ratio for a single acoustical element 100 is non-uniform. I.e. such that one region 116, 118 (shown in FIG. 5) of the acoustical element 100 is provided with predominantly the first fibre component 102 while another region 116, 118 of the acoustical element is provided with predominantly the second fibre component 104. As such, one single acoustical element 100 can be provided with differing acoustic properties over the surface

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thereof by varying the proportions of the first fibre component **102** and the second fibre component **104**.

The mixture **1** is then shaped **1006** into single layered tile shaped elements **2** whereby the mixture **1** is compressed with a compression ratio to a compressed state. The compression ratio may be controlled such that the compression ratio for a single acoustical element **100** is non-uniform. Increasing the compression ratio in a region of the acoustical element **100** provides increased airflow resistance and thus reduced sound absorbing qualities for that region, and thus increased sound reflection and possibly diffusion. Reducing the compression ratio in a region oppositely provides improved sound absorbing qualities for that region.

The shaping **1006** is preferably performed by means of a form pressing operation such as by means of vacuum forming, drape forming or pressure forming. Form pressing allows local variations of e.g. the thickness of the acoustical element **100** as well as the density/compression ratio thereof. I.e. such that different regions of the acoustical element **100** may have different thickness, topography and/or density/compression ratio, all of which allows providing different acoustical properties to different regions of the acoustical element **100**.

Moreover, the shaping **1006** may comprise a calendaring operation **1006a** in which the mixture **1** that is to form the single layered tile shaped element **2** and the acoustical element **100** is brought through calendaring rolls. The calendaring rolls may be smooth or provided with protruding patterns such that regions of the mixture **1** that is to form the acoustical element **100** receives a higher compression ratio than surrounding regions.

The method **1000** further comprises fixating **1008** the single layered tile shaped elements **2** in the compressed state for obtaining the acoustical elements **100**. The fixation **1008** of the single layered tile shaped elements **2** may be performed simultaneously as the shaping **1006**, e.g. during the form pressing operation, or as a separate subsequent manufacturing step. The fixating **1008** of the single layered tile shaped elements **2** comprises heating **1008a** the single layered tile shaped elements **2** in the compressed state for melting the thermoplastic outer layer of the bicomponent fibres and, additionally, for activating a separate binder. The second fibre component **104** and/or the binder is thus activated for fixating **1008** the single layered tile shaped elements **2** in the compressed state for obtaining the acoustical element **100**. The first and second fibre components **102**, **104** are consequently fixated **1008** in the desired shape of the acoustical element **100**.

The mixing **1004** the first fibre component **102** and the second fibre component **104** and shaping **1006** the mixture comprises controlling a mixing ratio of the first fibre component **102** in relation to the second fibre component **104** and/or the compression ratio of the mixture such that the acoustical elements **100** comprises a first group of the acoustical elements **100** having a first set of acoustic properties, such as sound absorbing properties, and a second group of the acoustical elements **100** having a second set of acoustic properties being different from the first set of acoustical properties.

As will be elaborated further on below, the method **1000** allows many variations of the acoustical elements **100** such that the manufacturing method **1000** allows provision of a series of acoustical elements **100** that are all tailored to suit their respective intended position and function in a room.

The method may further comprise arranging **1010a** core elements **10** in the mixture before the step of shaping **1006** of the mixture and removing **1010b** the core elements **10**

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from the mixture after the step of fixation **1008** of the mixture for formation of at least one cavity **106** in the acoustical element. This is illustrated in FIG. **1**, in which four core elements **10** provided to form four corresponding cavities **106** in an acoustical element **100**.

The core elements **10** are arranged **1010a** into their intended positions in the acoustical element **100** when it is still in the shape of a mixture **1** that is about to be shaped **1006** and fixated **1008** into its final shape as an acoustical element **100**. The core elements **10** are illustrated as cylindrical core elements **10**, but they could be realized with other shapes as well. For instance, the core elements **10** may be conically shaped, cuboid or pyramidal. However, as the core elements **10** preferably should be removable from the acoustical element **100** as the acoustical element **100** has been fixated **1008**, the core elements **10** must be arranged such that they can be accessed from one of the sides of the acoustical element **100** once it is fixated **1008**. It is however feasible that the core elements **10** may be configured to remain in the acoustical element **100** and thus not have to be removed therefrom, which would provide less constraints on how the core elements **10** has to be formed. For instance, the core elements **10** could in such an embodiment naturally be fully enclosed in the acoustical element **100** and thus not have to be accessible from any of the sides of the acoustical element **100** for removal.

When the core elements **10** are arranged in their respective intended position, the mixture **1** can be shaped **1006** and fixated **1008** into an acoustical element **100**.

The core elements **10** are, if desired, subsequently removed **1010b** after the single layered tile shaped element **2** is fixated **1008** into an acoustical element **100** and thus maintains its shape. The core elements **10** may thus generate cavities **106** in the acoustical element **100** that corresponds to the shape of the core elements **10**. Each cavity **106** provides acoustic properties to the acoustical element **100**, as each cavity **106** may alter the airflow resistance through the acoustical element **100** and/or resonance properties throughout the fibre structure. The cavities **106** may thus improve the sound absorption of the acoustical element **100** locally where the cavities **106** are arranged.

The cavities **106**, which may be formed by the core elements **10**, are shown being evenly distributed throughout the acoustical element **100**. The core elements may thus be arranged for formation of a plurality of evenly distributed cavities **106** in each acoustical element **100**. However, the core elements **10** may be arranged **1010a** unevenly distributed in the mixture **1** as well. For instance, it may be desired to provide an acoustical element **100** that gradually transitions from mainly reflecting sound waves to mainly absorbing sound waves or vice versa. In such an application, the arrangement **1010a** of core elements **10** can be performed accordingly such that more core elements **10**/cavities **106** are arranged in the region of the acoustical element **100** in which high sound absorption is desired and vice versa.

The method may be carried out in a piece-by-piece operation or in a line-based operation. The line-based operation may comprise one or more production lines. For instance, a first production line may be utilized for making acoustical elements with a first set of acoustic properties and a second production line may be utilized for making acoustic elements with a second set of acoustic properties.

In the following, the method **1000** illustrated in FIG. **1** will be further explained in relation to a number of different embodiments of finished acoustical elements **100**. It is to be realized that while some embodiments are separately illus-

trated and described, the various embodiments can be freely combined into a single acoustical element **100**.

FIG. **2** shows a perspective view of an acoustical element **100** in which the acoustical element is provided **1012** with a surface layer **108**. As is shown, the surface layer **108** may be provided **1012** on a front surface **112a** of the acoustical element **100**. The front surface **112a** is the surface intended to face the room in which the acoustical element **100** is mounted. However, a surface layer **108** may be provided **1012** also to an opposite rear surface **112b** of the acoustical element **100**.

The provision **1012** of the surface layer **108** may be performed before the shaping **1006** of the mixture **1**, as is illustrated in FIG. **1**, such that that the surface layer **108** is integrated into the acoustical element **100** during the shaping **1006** and fixation **1008** of the mixture **1**. However, the surface layer **108** could also be applied after the shaping **1006** or after the fixation **1008**.

The surface layer **108** may be provided with perforations **114** as is shown in FIG. **2**. The perforations **114** may be uniformly distributed over the surface layer **108** or, as is illustrated in FIG. **4**, be distributed in a non-uniform manner. This is one way to achieve a non-uniform air-flow resistance through the surface layer **108** which will follow the distribution of the perforations **114**. In a region of the acoustical element **100** in which there are many perforations **114**, less air-flow resistance is generated, and thus improved sound absorption compared to a region having less perforations **114**.

Moreover, the method **1000** may perform the steps of providing **1002** a first fibre component **102** in the form of mineral wool and a second fibre component **104** and mixing **1004** the first fibre component **102** and the second fibre component **104** for provision of a mixture **1** twice such that separate layers **110a**, **110b** are formed in the acoustical element **100**. Each layer **110a**, **110b** may be provided with different mixing ratios of the first fibre component **102** and the second fibre component **104**.

FIG. **3** schematically shows an acoustical element **100** according to one embodiment. In FIG. **3**, the first and second fibre components **102**, **104** in the mixture are shaped **1006** such that the main fibre direction is in the longitudinal direction **L** of the acoustical element **100**. The longitudinal direction **L** is herein defined as a direction being parallel with a main extension plane of the element **100**. The shaping **1006** may thus comprise a step of directionally arranging the first and second fibres **102**, **104** in a desired direction.

One example of how this may be achieved is by treating the mixture with a rotating spiked mandrel.

FIG. **3** illustrates the main general direction in which the fibres of the first and second fibre components **102**, **104** are to be arranged and is thus simplified, it is to be understood that individual fibres may deviate from the main direction.

FIG. **4** similarly to FIG. **3** schematically shows an acoustical element **100** according to another embodiment in which the main fibre direction of the first and second fibre components **102**, **104** are arranged in a transverse direction **T** which is essentially perpendicular in relation to the longitudinal direction **L** and extends in the thickness direction of the element.

The fibre length of the first and second fibre component may be configured to be smaller than an intended thickness of the acoustical element, whereby the fibres are free to be arranged in any direction. For example, the fibres may be oriented in the **T**-direction by forming the mixture into a web which is moved slower than an infeed rate of the fibre

mixture, resulting in a packing of the mixture which orientates the fibres in the desired transverse direction **T**.

FIG. **5** shows an acoustical element **100** in which the method **1000** comprises controlling the mixing ratio such that the mixing ratio for a single acoustical element **100** is non-uniform. As such, a first region **116** may be achieved in which the first fibre component **102** is present in higher or lower proportion in relation to the second fibre component **104** than it is in the remaining acoustical element **100**. Any number of such regions **116** may be provided, each exhibiting different acoustical properties depending on the controlled mixing ratio for each first region **116**.

Moreover, the method **1000** may comprise controlling the compression ratio such that the compression ratio for a single acoustical element **100** is non uniform. In FIG. **5**, the acoustical element **100** is provided with a second region **118** in which the compression ratio of the mixture may be higher or lower than the remaining acoustical element **100**. Increasing the compression ratio increases the airflow resistance and thus reduces the sound absorption. Each acoustical element **100** can thus be manufactured by the method **1000** with tailored acoustic properties. Any number of such regions **118** may be provided, each exhibiting different acoustical properties depending on the controlled compression ratio for each second region **118**.

FIG. **6** further discloses an acoustical element **100** which is manufactured by means of the method **1000** such that it has a non-uniform thickness. The step of shaping **1006** the mixture may thus comprise forming single layered tile shaped elements, acoustical elements **100**, with non-uniform thickness, such that the acoustical elements obtain a front and/or rear surface provided with a topography, i.e. a non planar surface. The shaping **1006** may be controlled to achieve the desired thickness by means of using a corresponding mould **3** (shown in FIG. **1**) into which the mixture **1** is arranged for shaping **1006** and fixating **1008**. Variations in thickness of the acoustical element **100** affects the sound absorption properties of the acoustical element **100**, for instance will an increased thickness for most materials provide improved absorption for lower sound frequencies while the opposite is true for a reduced thickness.

FIG. **7** shows a perspective view of an acoustical element **100** for which the shaping **1006** of the mixture into single layered tile shaped elements **2**, to form acoustical elements **100**, is performed such that the single layered tile shaped elements **2** are provided with a single curved or double curved extension which is then fixated **1008** into corresponding acoustical elements **100**. The acoustical element **100** shown is provided with a single curved extension, the curvature may be arranged in the longitudinal direction **L** or in the transverse direction **T** of the room in which the acoustical element **100** is mounted. A double curved extension is however equally possible in which the curvature is in both the longitudinal and transverse directions **L**, **T**. The convex curvature of the front surface **112a** of the acoustical element **100** facilitates diffusion of sound.

FIG. **8** shows an acoustical element **100** for which the method **1000** comprises shaping **1006** of the mixture **1** into single layered tile shaped elements **2** such that the single layered tile shaped elements **2** obtain a front surface **112a** and/or rear surface **112b** provided with a single curved or double curved topography. The single layered tile shaped elements **2** are fixated **1008** to form corresponding acoustical elements **100**. The acoustical element **100** shown is provided with a single curved topography, the curvature may be arranged in the longitudinal direction **L** or in the transverse direction **T** of the room in which the acoustical

element **100** is mounted. A double curved topography is however equally possible in which the curvature is in both the longitudinal and transverse directions L, T. The convex curvature of the front surface **112a** of the acoustical element **100** facilitates diffusion of sound

FIG. **9** further shows a suspended ceiling system **200** comprising ceiling tiles in the form of acoustical elements **100** manufactured by the method **1000** according to the teachings herein, the ceiling tiles are obtained from the first and second groups **100a**, **100b**, **100c** with mutually different acoustic properties. The suspended ceiling system **200** may be suspended by means of a grid of profiles or in other ways as is realized by one skilled in the art.

For a suspended ceiling system **200** to provide acoustic properties that facilitates transmission of speech, such as in an auditorium or in a conference room or similar, it is typically desired that some sound waves are reflected back towards the speaker while others are reflected towards the audience. The sound waves reflected back towards the speaker should provide early reflections, i.e. be reflected back within 30 ms to 50 ms from when they are generated. Such early reflections should also be provided to the audience. Early reflections generally improve both speaker comfort and speech intelligibility. Reflections and reverberations which occurs later than 30 ms to 50 ms are however considered detrimental and should thus be reduced.

In such a context, the suspended ceiling system **200** may be provided with ceiling tiles in the form of acoustical elements **100** from a first group **100a** which comprises acoustical elements **100** having high sound absorption. These may, as is illustrated in FIG. **9**, be arranged behind a speaker position **202** and also towards the rear of the suspended ceiling system **200** behind an audience position **204**. This first group **100a** of ceiling tiles **100** will thus prevent sound reflections where it is not desired, i.e. from behind the speaker position **202** and behind the audience position **204**. The acoustical elements **100** in the first group **100a** may thus be manufactured according to the method **1000** disclosed herein with a reduced compression ratio and/or be provided with cavities etc. to achieve high sound absorption.

Moreover, ceiling tiles in form of acoustical elements **100** in a second group **100b** may be arranged above and slightly in front of the speaker position **202**. In this region of the suspended ceiling system **200**, it is desired to provide reflection and diffusion of sound. Preferably such that sound reaches both the speaker position **202** and the audience position **204**. As such, the acoustical elements **100** in the second group **100b** may be manufactured according to the method **1000** with an extension and/or topography that is single curved or double curved as is shown in FIGS. **7** and **8**. The curvature of the front surface **112a** of the acoustical elements in the second group **100b** arranged in the longitudinal direction L of the room facilitates sound diffusion and directionality as described above.

The suspended ceiling system **200** may further be provided with a third group **100c** of acoustical elements **100**, which are arranged between the speaker position **202** and at least partially above the audience position **204**. The third group **100c** being arranged farther away from the speaker position **202** than the second group **100b**. In this region of the suspended ceiling system **200**, reflections that are directed back to the speaker position **202** may be undesired as they would be detrimental to speaker comfort. As such, the acoustical elements **100** of the third group **100c** may be provided with acoustical properties that facilitates reflection but not wide angle diffusion of sound. The sound will thus

only be reflected towards the audience position **204** and not back to the speaker position **202**. Such acoustical elements **100c** of the third group could be manufactured by the method **1000** herein by for instance shaping **1006** the mixture **1** such that is essentially flat but with a relatively high compression ratio, to name one example.

It is to be realized that FIG. **9** illustrates but one example of a suspended ceiling system **200**, many variations are possible within the scope of the present disclosure. For instance, any number of different groups **100a**, **100b**, **100c** may be provided depending on the application. Manufacturing acoustical elements **100** according to the method **1000** described above allows each suspended ceiling system **200** to be delivered tailored to the intended use, while reducing the environmental impact as recycled fibre materials are used at least in the first fibre component **102**.

It will be appreciated that the present invention is not limited to the embodiments shown. Several modifications and variations are thus conceivable within the scope of the invention which thus is exclusively defined by the appended claims.

The invention claimed is:

1. A method for manufacturing acoustical elements, comprising:

providing a first fibre component in a form of mineral wool and a second fibre component in a form of bicomponent fibres having a core with a thermoplastic outer layer;

mixing the first fibre component and the second fibre component to provide a mixture;

shaping the mixture into a group of single layered tile shaped elements whereby the mixture is compressed with a compression ratio to a compressed state; and

fixating the group of single layered tile shaped elements in the compressed state for obtaining the acoustical elements, wherein

fixating the group of single layered tile shaped elements comprises heating the group of single layered tile shaped element in the compressed state for melting the thermoplastic outer layer of the bicomponent fibres, whereby the second fibre component is activated for fixating the group of single layered tile shaped elements in the compressed state,

mixing the first fibre component and the second fibre component and shaping the mixture comprises controlling a mixing ratio of the first fibre component in relation to at least one of the second fibre component and the compression ratio of the mixture for providing a first group of the acoustical elements having a first set of acoustic properties and a second group of the acoustical elements having a second set of acoustic properties being different from the first set of acoustical properties, and

the compression ratio is further controlled such that the compression ratio for a single acoustical element of the acoustical elements is non-uniform.

2. The method for manufacturing acoustical elements according to claim **1**, wherein shaping the mixture further comprises forming the group of single layered tile shaped elements in a form pressing operation.

3. The method for manufacturing acoustical elements according to claim **1**, wherein shaping the mixture further comprises forming the single layered tile shaped elements in a form pressing operation.

4. The method for manufacturing acoustical elements according to claim **1**, wherein the mixing ratio is further controlled such that a mixing ratio for a single acoustical

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element of the group of single layered tile shaped elements of the acoustical elements is non-uniform.

5 **5.** The method for manufacturing acoustical elements according to claim **1**, wherein shaping the mixture further comprises forming the single layered tile shaped elements of the acoustical elements with a non-uniform thickness.

6. The method for manufacturing acoustical elements according to claim **1**, further comprising

arranging core elements in the mixture before shaping of the mixture, and

removing the core elements from the single layered tile shaped elements of the acoustical elements after fixation of the group of single layered tile shaped elements to form cavities in the acoustical elements.

15 **7.** The method for manufacturing acoustical elements according to claim **6**, wherein the core elements are arranged to form a plurality of evenly distributed cavities in each acoustical element of the acoustical elements.

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8. The method for manufacturing acoustical elements according to claim **1**, wherein the shaping of the mixture into single layered tile shaped elements is performed such that the tile shaped elements are provided with a single curved or double curved extension.

9. The method for manufacturing acoustical elements according to claim **1**, wherein the shaping of the mixture into the group of single layered tile shaped elements is performed such that the single layered tile shaped elements obtain at least one of a front surface and a rear surface provided with at least one single curved or double curved topography.

10. The method for manufacturing acoustical elements according to claim **1**, further comprising providing each acoustical element of the acoustical elements with a surface layer forming at least one of a front and back layer thereof.

15 **11.** The method for manufacturing acoustical elements according to claim **10**, wherein the surface layer has non-uniform air flow resistance.

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